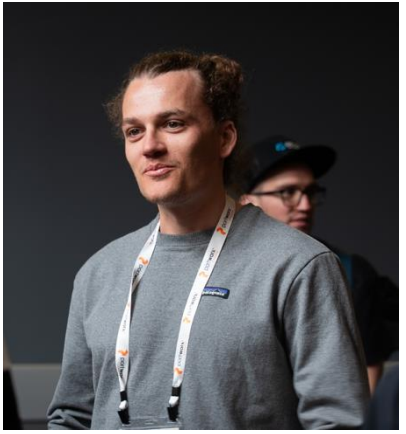
swisscom

The Accidental Platform Team: Kubernetes Operators at Swisscom

21. Octobre 2026, Lea Brühwiler, Fabian Schulz





Fabian Schulz

DevOps Engineer

fabian.schulz1@swisscom.com



Lea Brühwiler

DevOps Engineer

lea.bruehwiler@swisscom.com


swisscom



User Equipment

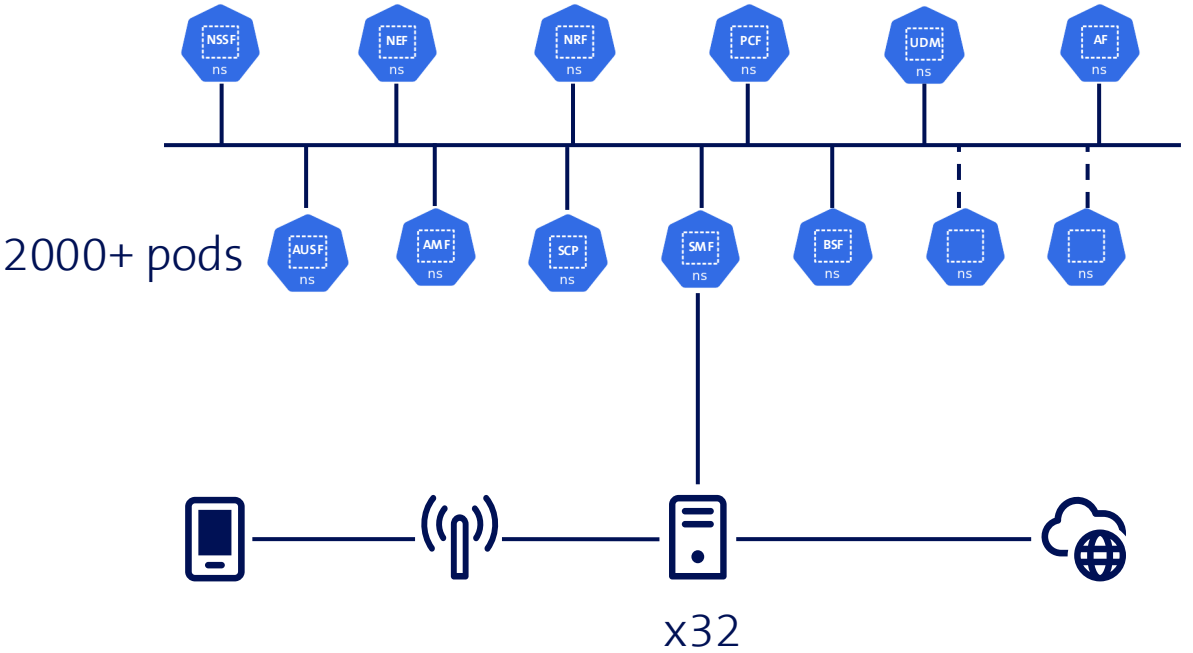
(e.g. Phone, IoT Device)

Radio Access Network

Mobile Data Core

Data Network

(e.g. Internet)


NSSF
ns
NEF
ns
NRF
ns
PCF
ns
UDM
ns
AF
ns
2000+ pods
AUSF
ns
AMF
ns
SCP
ns
SMF
ns
BSF
ns
ns
ns
x32


DC
DC
DC


Network Setup Wizard
Welcome to the Network Setup Wizard
This wizard will help you set up this computer to run on your network. With a network you can:
• Share an Internet connection
• Set up Internet Connection Firewall
• Share files and folders
• Share a printer
To continue, click Next.
< Back
Next >
Cancel


Yer a wizard, Harry.







routers

5G core deployment

Kubernetes

OS

Server Hardware

routers





Create IP assignments

2 days

Collect IP in IPAM

2 days

Populate inventory

1 day

Create change requests

5-10 days

MetalLB

k8s /CNF

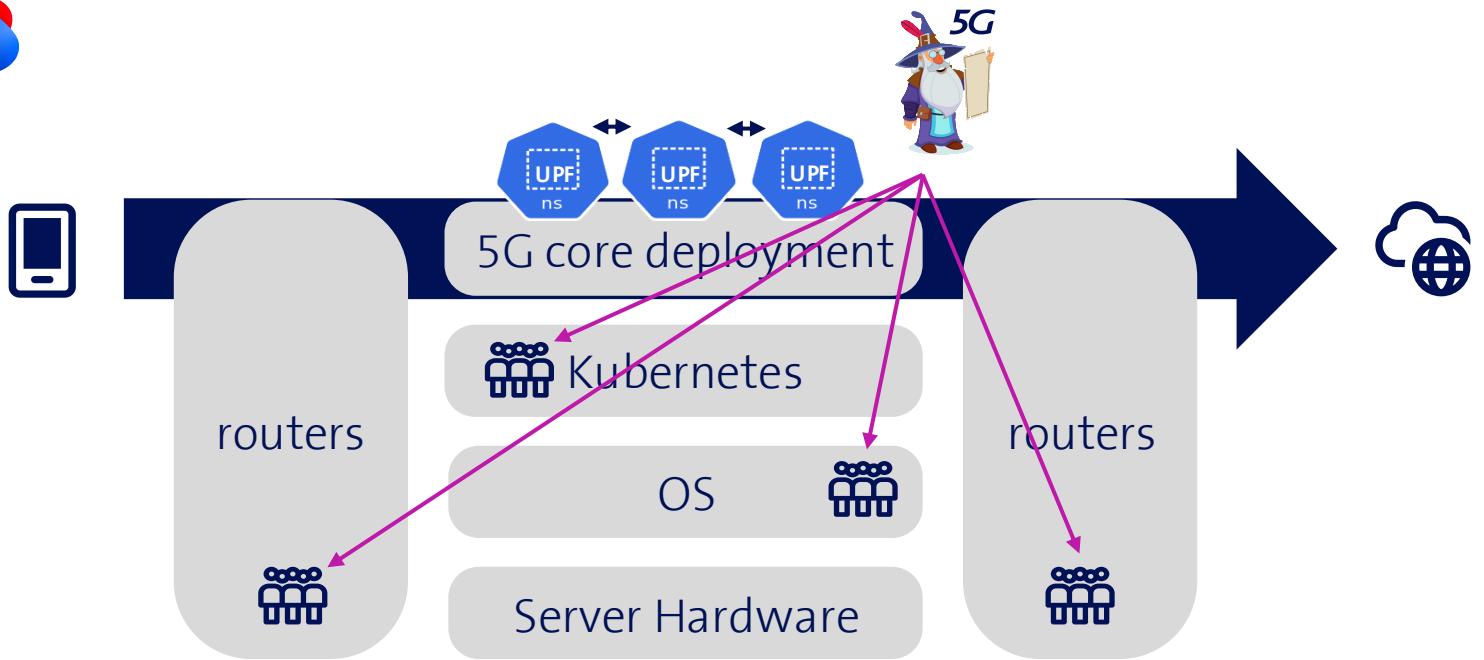
Thor UI

edge router

Integrate, test and observe

5 - 10 days

Landslide


5G
UPF
ns
UPF
ns
UPF
ns
5G core deployment
routers
Kubernetes
OS
Server Hardware
routers

Easy, right?



Pain Points When Deploying a 5G Core Network



Cross-domain Knowledge

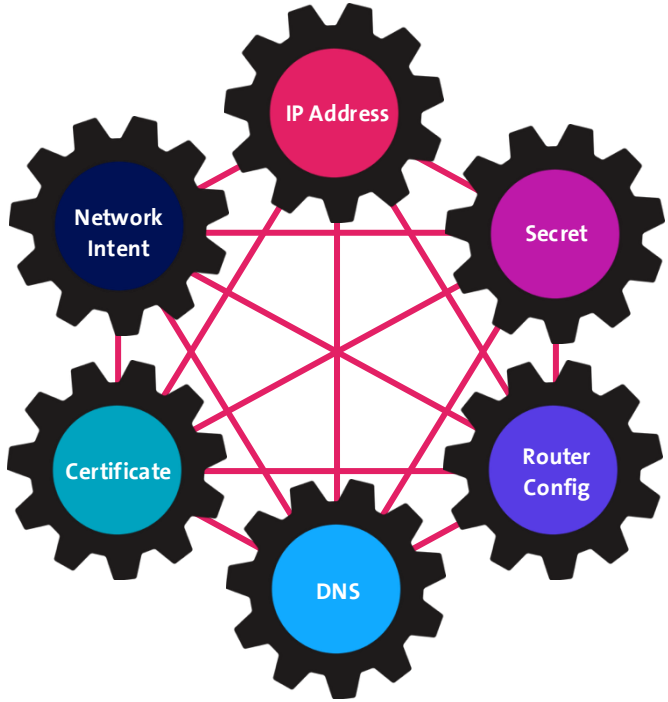


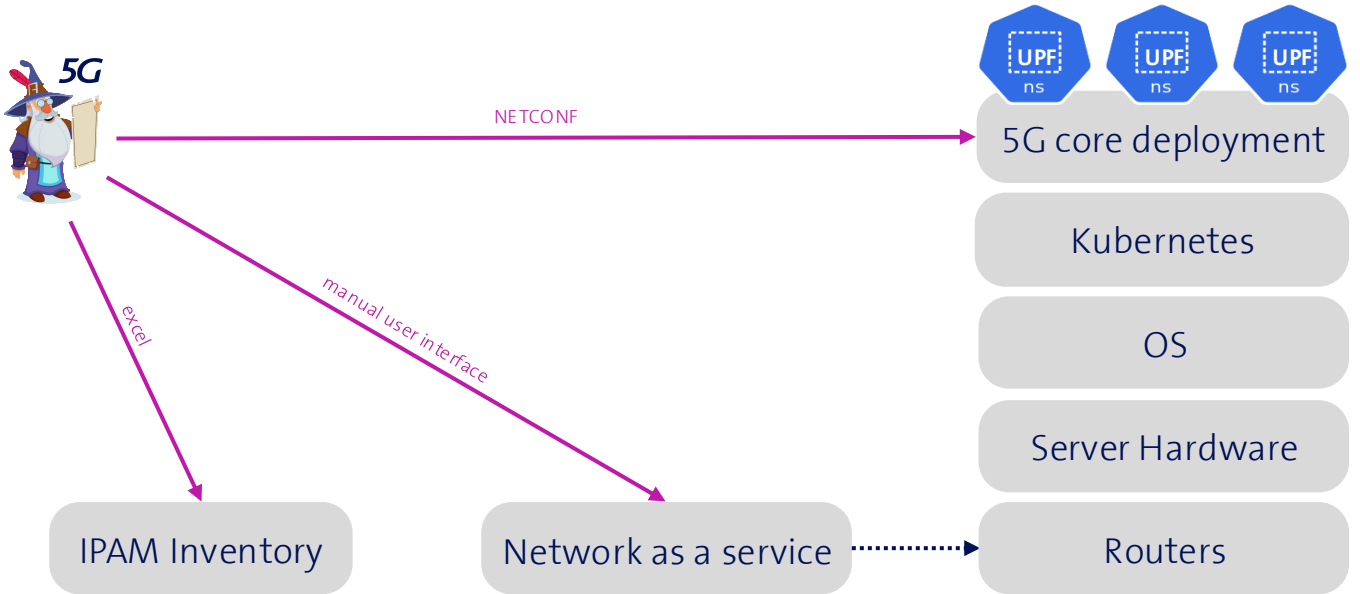
Manual Process

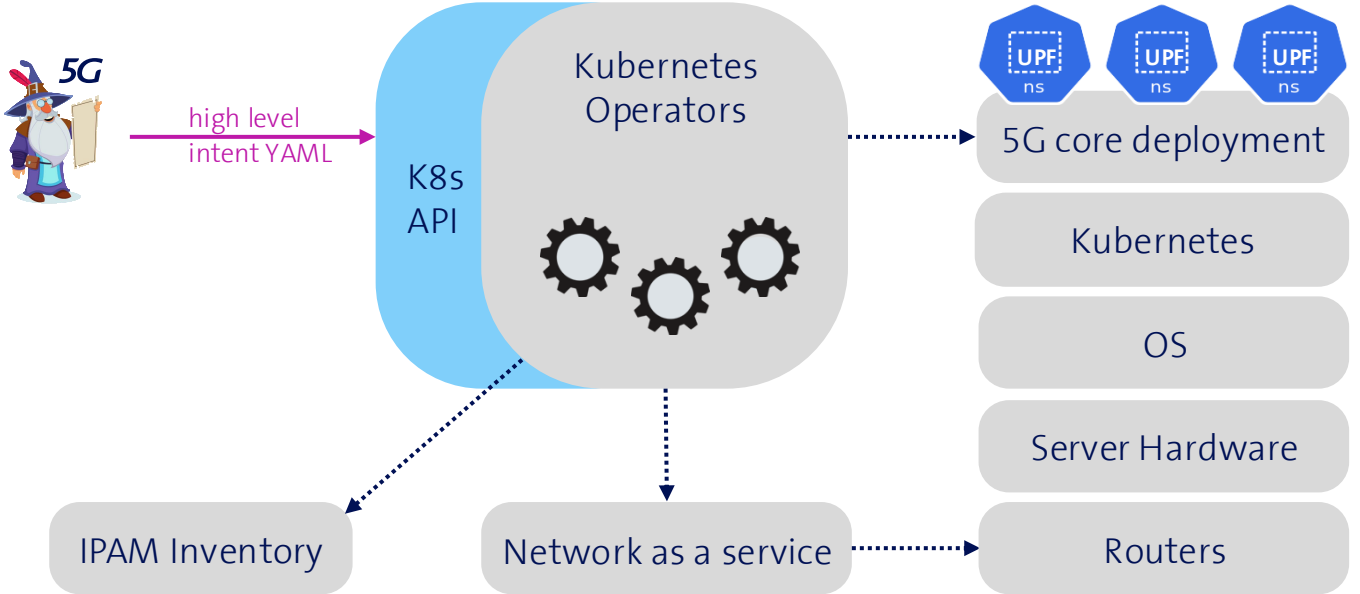


Static Config


high
level
network
intent
5G


IP Address
Network Intent
Secret
Certificate
Router Config
DNS


5G
NETCONF
UPF
ns
UPF
ns
UPF
ns
5G core deployment
Kubernetes
OS
Server Hardware
Routers
excel
manual user interface
IPAM Inventory
Network as a service


5G
high level
intent YAML
K8s
API
Kubernetes
Operators
UPF
ns
UPF
ns
UPF
ns
5G core deployment
Kubernetes
OS
Server Hardware
Routers
IPAM Inventory
Network as a service



Cross-domain Knowledge



Manual Process



Static Config

Domain Knowledge in Operators

Automation

Constant Reconciliation

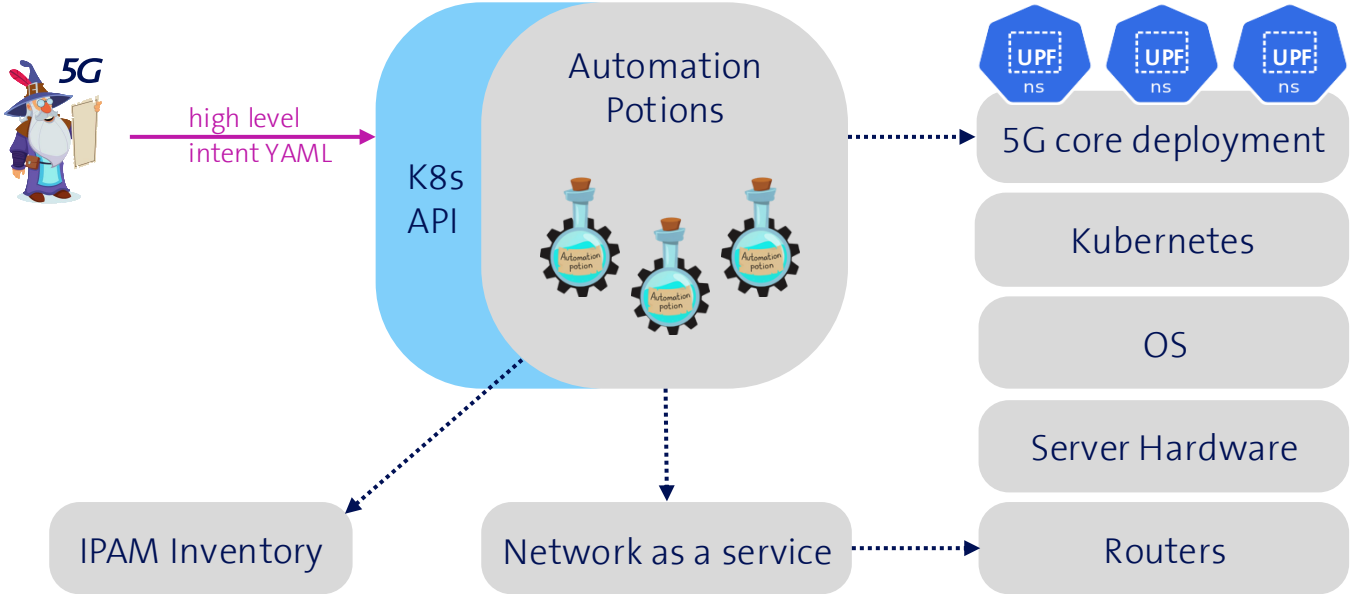
More details?



https://architecture.cncf.io/architectures/swisscom-cloud-native-telco

How brave do you feel?




5G
high level
intent YAML
K8s
API
Automation
Potions
UPF
ns
UPF
ns
UPF
ns
5G core deployment
Kubernetes
OS
Server Hardware
Routers
IPAM Inventory
Network as a service



Gradual Approach



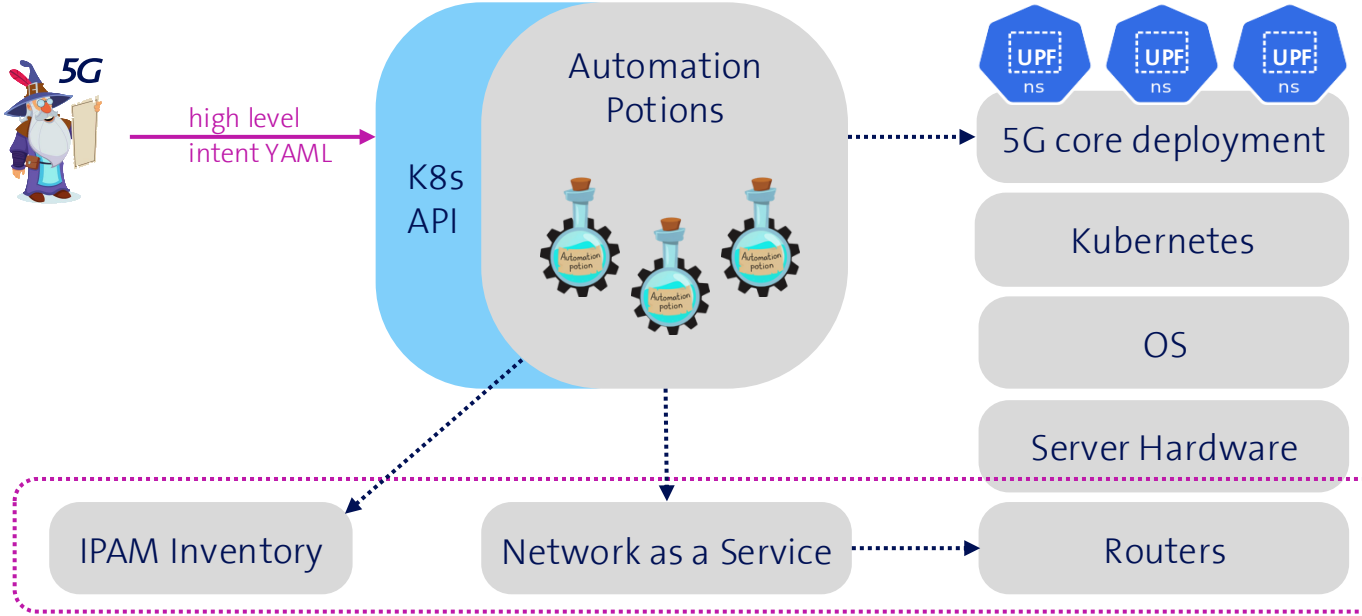
Select first easy use case



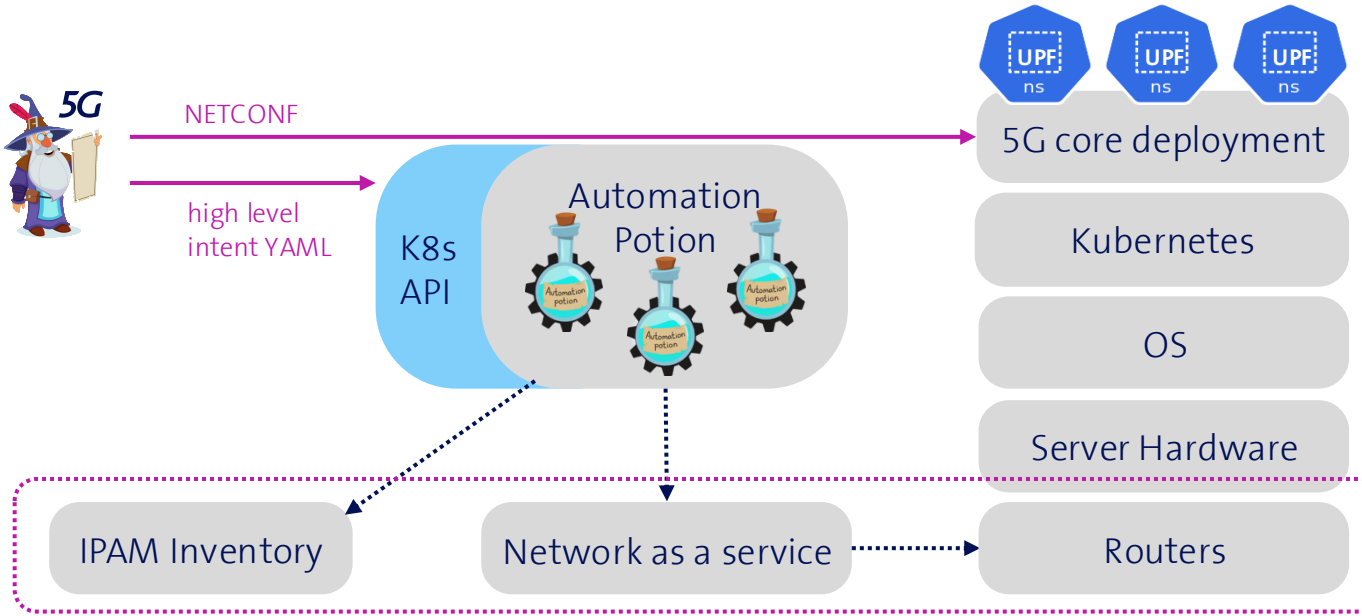
Ship first prototype early



Iterate on feedback

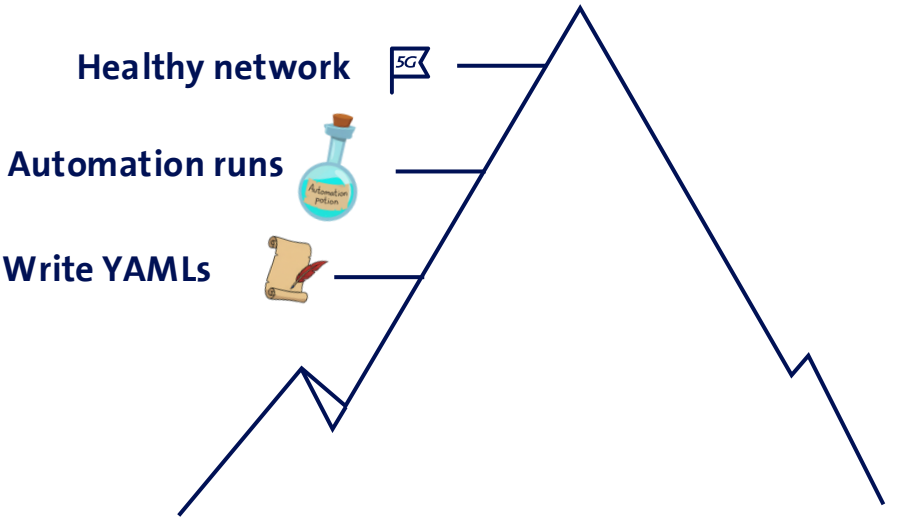


First Use Case



First Use Case


5G


Healthy network
5G
Automation runs
Automation potion
Write YAMLs

Let’s goooo



We are done, right?





```
// CcdSecondaryNetworkTopologySpec defines the desired state of CcdSecondaryNetworkTo
// +kubebuilder:validation:XValidation:rule="(!self.cpConfig.topology.contains('Bgp')
type CcdSecondaryNetworkTopologySpec struct {
        // Configuration needed for IPAM in Netbox
        NetBoxConfig NetBoxConfigTopology `json:"netBoxConfig,omitempty"`

        // Configuration needed for management of ConnectionPoints in CPBroker
        CpConfig CpConfig `json:"cpConfig,omitempty"`

        // InternalNetwork is used to determine if the network is internal or externa
        // If set to true, the CP will be created with createRe
        // Default: false
        // Immutable after creation.
        // +default:value=false
        // +kubebuilder:validation:Optional
        // +kubebuilder:validation:XValidation:rule="self == ol
        InternalNetwork bool `json:"internalNetwork,omitempty"`

        // Interval for MetalLB BFDProfile receiveInterval, Met
        // Must be between 10 and 3000
        // Default: 300
        // +default:value=300
        // +kubebuilder:validation:Minimum=10
        // +kubebuilder:validation:Maximum=3000
        BfdInterval int32 `json:"bfdInterval,omitempty"`

        // Determines the number of consecutive missed BFD pack
        // Must be between 2 and 10
        // Default: 4
        // +default:value=4
        // +kubebuilder:validation:Minimum=2
        // +kubebuilder:validation:Maximum=10
        BfdMultiplier int32 `json:"bfdMultiplier,omitempty"`

        // Defines the ASN used on the CNF side.
        // This field is required and immutable once set
        // Required
        // Example: "65002"
        // +kubebuilder:validation:XValidation:rule="self == oldSelf",message="Field
        CnfAsn string `json:"cnfAsn,omitempty"`

        // Configuration of the git repository for the fabric information export
        // Default: {}
        // +kubebuilder:default:={}
        FabricOutputConfig FabricOutputConfig `json:"fabricOutputConfig"`
}
```

```
// +kubebuilder:validation:XValidation:rule="(self.ipv6.enable || self.ipv4.enable)"
type NetBoxConfigTopology struct {
        // NetBox Tenant that will be used for creating NetBox IPAM Resources
        // Example: "SGDMC"
        // Required
        // Immutable after creation.
        // +kubebuilder:validation:Required
        // +kubebuilder:validation:XValidation:rule="self == oldSelf",message="Field 'tenant' is immutable"
        Tenant string `json:"tenant,omitempty"`
        // Indicating if we are selecting IPv6 IPs, and the selectors used
        // Required

                                                                    quired
                                                                    lidation:rule="self == oldSelf",message="Field 'ipv6' is immutable"
                                                                     `json:"ipv6,omitempty"`
                                                                    cing IPv6 IPs, and the selectors used


                                                                    quired
                                                                    lidation:rule="self == oldSelf",message="Field 'ipv4' is immutable"
                                                                     `json:"ipv4,omitempty"`



                                                                    IPv6 selection criteria.
                                                                    ule="(self.enable == false || (has(self.parentPrefixSelector) && has(self.pref


                                                                    ned


                                                                    in NetBox Operator. Note that the "family" field will be added by the Controll
                                                                     family (IPv4/IPv6)

                                                                    dation:rule="self == oldSelf",message="Field 'parentPrefixSelector' is immutabl
        parentPrefixSelector map[string]string `json:"parentPrefixSelector,omitempty"`

        // The desired prefix length for the IPv6 prefix. Be aware that this is an integer which will be converted to
        // string in slash notation for NetBox Operator
        // Immutable after creation.
        // Must be between 8 and 128
        // Example: 64
        // +kubebuilder:validation:Minimum=8
        // +kubebuilder:validation:Maximum=128
        // +kubebuilder:validation:XValidation:rule="self == oldSelf",message="Field 'prefixLength' is immutable"
        PrefixLength int `json:"prefixLength,omitempty"`
}
```

Maybe not quite:

- We don't have time to test your stuff.
- You cannot deploy a 5G Core with GitOps because there will always be manual changes on the box.
- How can I see which changes will be applied to the cluster?
- We had a UI which was more intuitive!
- How can I be sure it worked?
- I think we were faster the old way.
- How do we know where this change is coming from?
- I don't understand Kubernetes.
- I don't even know how to start using it.
- I just did it myself using a Gitlab pipeline, it's easier that way.

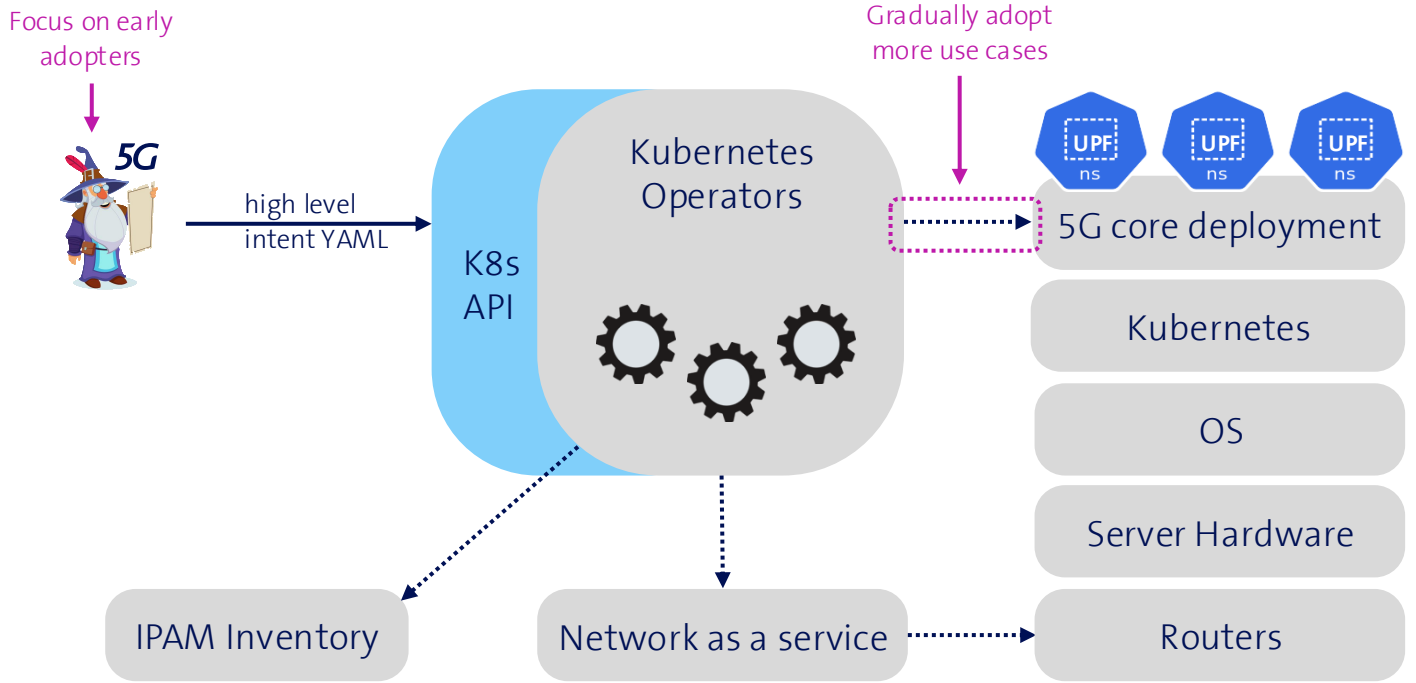


- We don’t have time to test your stuff.
- You cannot deploy a 5G Core with GitOps because there will always be manual changes on the box.

Gradual Adoption



How do we get our solution tested?



- We don’t have time to test your stuff.
- You cannot deploy a 5G Core with GitOps because there will always be manual changes on the box.

Gradual Adoption

- How can I see which changes will be applied to the cluster?
- We had a UI which was more intuitive!
- How can I be sure it worked?
- I think we were faster the old way.
- How do we know where this change is coming from?
- I don’t understand Kubernetes.
- I don’t even know how to start using it.
- I just did it myself using a Gitlab pipeline, it’s easier that way.

- How can I see which changes will be applied to the cluster?
- We had a UI which was more intuitive!

Graphical UI





- We don’t have time to test your stuff.
- You cannot deploy a 5G Core with GitOps because there will always be manual changes on the box.

> **Gradual Adoption**



- How can I see which changes will be applied to the cluster?
- We had a UI which was more intuitive!

> **Graphical UI**



- How can I be sure it worked?
- I think we were faster the old way.
- How do we know where this change is coming from?

- I don’t understand Kubernetes.
- I don’t even know how to start using it.
- I just did it myself using a Gitlab pipeline, it’s easier that way.

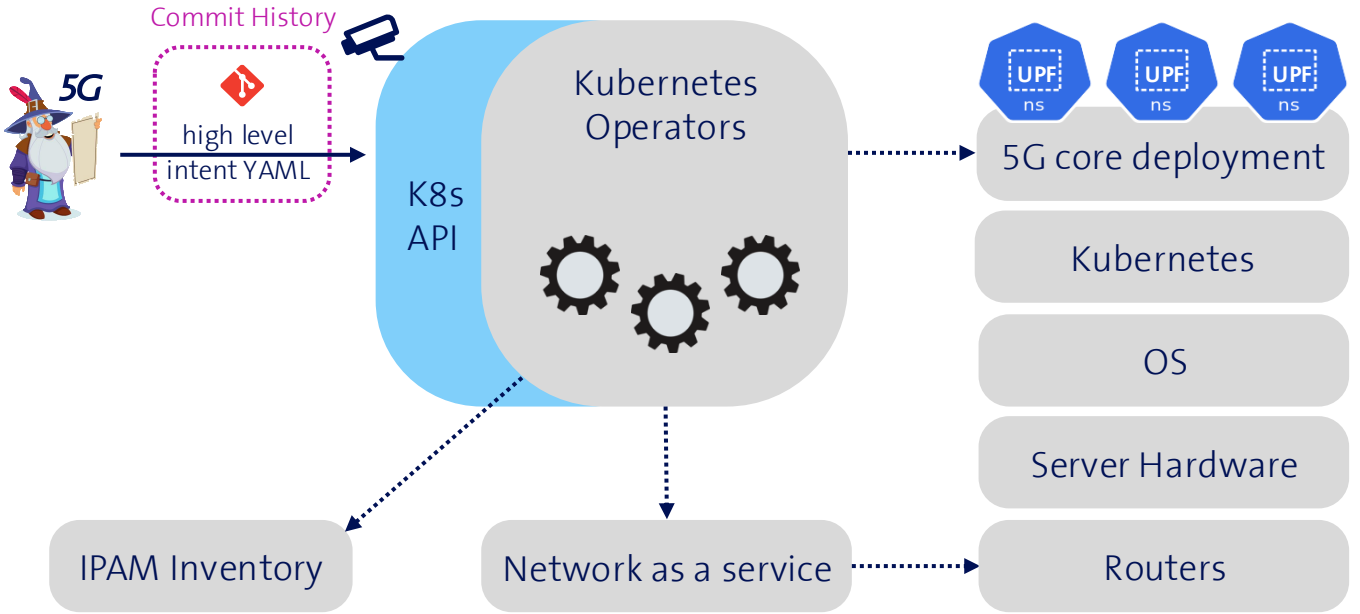
- How can I be sure it worked?
- I think we were faster the old way.
- How do we know where this change is coming from?

Observability and Traceability





How do we know what happened?




Home › Dashboards › ROPEs Production
Add
Share
Last 6 hours
Tenants
5G
Total Connection Points
77
Total Static Routes
144
Prefix Claims
491
IP Range Claims
134
Primary Networks
32
Secondary Networks
27
IP Address Claims
205
MetalLB IP Pools
169
Network Instances
6
Network Instances
6
Network Instance
Network Instance
Status
Ready
Network Instance
Network Instance
Status
Ready
Primary Networks
Primary Network Na
Status
Ready
Secondary Networks
Secondary Network
Status
Ready
Static Routes
Static Route Name
Status
Ready

- We don’t have time to test your stuff.
- You cannot deploy a 5G Core with GitOps because there will always be manual changes on the box.

> **Gradual Adoption**



- How can I see which changes will be applied to the cluster?
- We had a UI which was more intuitive!

> **Graphical UI**



- How can I be sure it worked?
- I think we were faster the old way.
- How do we know where this change is coming from?

> **Observability and Traceability**



- I don’t understand Kubernetes.
- I don’t even know how to start using it.
- I just did it myself using a Gitlab pipeline, it’s easier that way.

- I don’t understand Kubernetes.
- I don’t even know how to start using it.
- I just did it myself using a Gitlab pipeline, it’s easier that way.

Knowledge Sharing

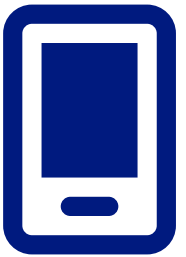


Establish Trust with Users

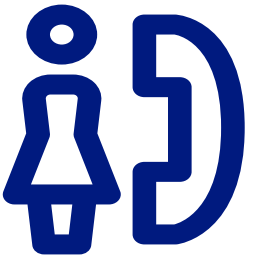
Design / Feedback Sessions



Focused Communication Channels



Regular Check-ins



Knowledge-Sharing Initiatives





- We don’t have time to test your stuff.
- You cannot deploy a 5G Core with GitOps because there will always be manual changes on the box.

→ **Gradual Adoption**



- How can I see which changes will be applied to the cluster?
- We had a UI which was more intuitive!

→ **Graphical UI**



- How can I be sure it worked?
- I think we were faster the old way.
- How do we know where this change is coming from?

→ **Observability and Traceability**



- I don’t understand Kubernetes.
- I don’t even know how to start using it.
- I just did it myself using a Gitlab pipeline, it’s easier that way.

→ **Knowledge Sharing**



The Tale of the Accidental Platform Team



Does it actually work?



Create IP tables

2 days

Collect IP in IPAM

2 days

Populate inventory

1 day

Create change requests

5-10 days

Integrate, test and observe

5 - 10 days

Create high-level intent

2 hours

Create lower-level intents

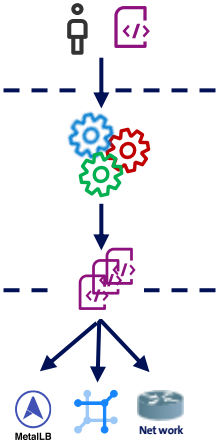
5 seconds

Realize intents / create CRs

5-10 minutes

Integrate, test and observe

2 hours and continuous testing & monitoring



And how brave do you feel now?



Thanks to colleagues for what we've achieved **together** in the past **1.5 years**.

We just created all endpoints **without a single error!**



I'm convinced that we've **saved a week or two of toil**.

Now, it took me **less than one hour** to create those networks!

What does the future hold?

Automation of all steps

What do we do regarding manual steps?

Different types of systems

How about imperative systems?

AI

Where can we meaningfully use it?



How to survive as an Accidental Platform Team?





How We Are Moving from GitOps to Kubernetes Resource Model in 5G Core

KubeCon Europe

2024

Just Claim It: Simplifying Network Automation with NetBox Operator

Cloud Native Telco Day

2025

Why Is It So Hard to Run a 5G Core on Kubernetes—And What Needs to Change for 6G

KubeCon + CloudNativeCon Europe

2026

2024

Building and Operating a Highly Reliable Cloud Native DNS Service With Open-Source Technologies

Container Days Conference

2025

The Cloud-Native Advantage: Intent-based Network Automation (SDC)

Container Days Conference

Thank you!





Wizard image credits: Nikola Malić

github.com/swisscom/cloud-native-telco